# Transformation in ETL Assignment  Questions

## Question 1: Define Data Transformation in ETL and explain why it is important.

**Answer**

Data transformation comes from (ETL) transformation means the raw, unorganized data convert into organized, clean, structured data analysis ready.

Its applying sets of business rules, calculation cleaning techniques restructuring method ensure the data is clean and ready to decision making.

**Importance:** It ensures consistency, accuracy, and compatibility across systems, making data reliable for analytics, reporting, and decision-making.

## Question 2 : List any four common activities involved in Data Cleaning.

**Answer**

### 1. Remove missing values

- Delete missing rows when % of missing rows is very small
- Replace missing values with mean or median when data is numerical and with mode when there is categorical data.
- For Time series use either forward fill or backward fill

2. **Remove duplicate values** = eliminates repeated records.

3. **Correct inaccurate values**

4. **Correct inconsistencies formatting**= Standardizing formats (e.g., dates, categories).

5. **Outlier detection** – identifying and treating extreme values.
    - Z-Score Method
    - IQR Method (Interquartile Range)

Question 3 : What is the difference between Normalization and Standardization?

**Answer**

- Data Normalization is the process of scaling numerical values so that all features fall within a uniform and comparable range.

**Min–Max Normalization (0 to 1 scaling)**

- Data standardization Converts data to have mean = 0 and standard deviation = 1 (z-score).

**Z-Score Standardization (mean = 0, SD = 1)**

**Key Difference:** Normalization is range-based, while Standardization is distribution-based.

Question 4 : A dataset has missing values in the “Age” column. Suggest two techniques to handle this and explain when they should be used.

**Answer**

**Mean/Median Imputation:**

Use when missing values are random and the dataset is large.

Median is preferred if the age distribution is skewed.

**Deleting rows:**

Use when missing values are very few (e.g., <5%) and removal won’t affect analysis.

Question 5: Convert the following inconsistent “Gender” entries into a standardized format (“Male”, “Female”): ["M", "male", "F", "Female", "MALE", "f"]

**Answer**

Original: ["M", "male", "F", "Female", "MALE", "f"]

Standardized: ["Male", "Male", "Female", "Female", "Male", "Female"]

Question 6 : What is One-Hot Encoding? Give an example with the categories: “Red, Blue, Green”.

**Answer**

**One-Hot Encoding** converts categorical values into binary columns (0/1).

| Categories | RED | BLUE | GREEN |
|---|---|---|---|
| RED | 1 | 0 | 0 |
| BLUE | 0 | 1 | 0 |
| GREEN | 0 | 0 | 1 |

Question 7 : Explain the difference between Data Integration and Data Mapping in ETL.

**Answer**

**Data mapping** defines how fields from one dataset correspond to fields in another dataset.

Helps in merging, integrating, and maintaining consistency.

**Data Integration** Combining data from multiple sources into a unified view.

Integration is about merging datasets, while Mapping is about aligning schema/fields.

Question 8: Explain why Z-score Standardization is preferred over Min-Max Scaling when outliers exist.

**Answer**

Min–Max Normalization (0 to 1 scaling)

- Converts data to a fixed range (usually 0 to 1)
- Maintains the shape of the original distribution
- Sensitive to outliers → values far from the rest can distort scaling

- Compresses data into [0,1], but outliers can distort the range, making most values cluster tightly.

Z-Score Standardization (mean = 0, SD = 1)

- Converts values into how many standard deviations they are from the mean
- Normalized data will have:
- Uses mean and standard deviation, reducing the impact of outliers by centering data around 0 with unit variance.

  Its preferred because Z-score is more robust(strong) when outliers are present.